Paper Review: Role- and Persona-Conditioned LLM Agents in Multi-Turn Dialogue Analysis (OPTION -1)

Topic: How role- and persona-variant LLMs operate in multi-turn dialogue systems
Readings: [1] NewsInterview (ACL 2025) and [2] ChatDev (ACL 2024)

1. **Problem Statement & Contribution**

Both papers test whether, when LLM agents are given a role or persona, they can carry out goal-directed, multi-turn dialogue the way people operate in those roles. NewsInterview[1] has the model (foundation or pre-trained) ask the next interview question and extract information over a full conversation after being given starting dialogues and a persona for the source (clueless, anxious, angry), and output via next questions or fact extraction from the source. ChatDev[2] has agents playing different roles like CTO, CEO, Programmer, Tester, and builds software from a one-line input such as “Build a specific game”, making them talk in pairs until they produce a piece of final code. Behaviour is set using the system prompt. LLMs use less grounding language like acknowledgements and confirmations than people do (Shaikh et al.,2024), and multi-turn dialogue data is scarce due to confidentiality.

[2] shows how role-playing agents can be set up to finish a task by chatting, i.e., by a fixed chain of two-agent chats that decide what they talk about and ask specific questions before answering. Unlike MetaGPT, whose agents follow fixed human-written procedures, its agents refine their solutions through dialogue. Its novelty lies mainly in how chats are organized. Where [2] organizes conversation between agents, [1] asks how well an LLM conducts conversation itself, using interviews. It finds that LLM-generated interview questions(versus the provided ones) rarely acknowledge the answers and don't steer the conversation over many turns. Instead of crowdsourced data, it contributes around 45k interviews to measure this over a whole conversation, though this high-quality dataset only covers 50 samples to detect the speaker role.

2. **Approach Details**

NewsInterview filters 487,310 transcripts with Llama-3.1-70B to 45,848 two-person interviews ,labelling as interviewe the speaker with more question marks. In the analysis, an LLM sees the first t-1 turns and writes the next question under 4 prompts(baseline, CoT, outline, outline+ CoT).GPT-4o rates match on six dimensions, and questions get discourse roles such as follow-up, acknowledgement, or topic change. In the game, the interviewer asks, and the source finds the persuasion number (1-5) to calculate how many facts to reveal nd replies in character. Reward is distinct facts revealed per answer.ChatDev chat chain runs design, coding, and testing as five subtasks. Each is a pairwise A-B chat until agreement and passes the result forward. Full history stays with a phase; only each phase’s final result carries on. Introduced Communicative Dehallucination to avoid vague results and response generation. The role is again defined using system prompts as similar to NewsInterview[1].

Disclosure in [1] by the agent depends on how persuaded it feels over the course of the whole interview, so building trust early pays off later. The eight source personas come from a journalism textbook(Sedorkin, 2015), and each of them needs a different approach. In [2], agents talk in pairs, and only each phase’s final result is passed to the next phase. Both designs rest on assumptions. [1] assumes LLM can tell how persuaded it feels, but its rating only moderately matched real people’s (r= 0.43). [2] assumes that the agents decide on their own when they agree. The shared trade-off is cost : [2] takes about 9.5 times as long and 3 times as many tokens as a single agent, and vague requests still produce basic programs.

3. **Evidence & Findings**

Evaluation setup. NewsInterview's analysis has Llama-3.1-70B write the next question under four prompts and compares the results with a former professional journalist doing the same task (Table 1 in [1]). Its game runs 450 interviews with GPT-4o, GPT-4o-mini, Llama-3.1-70B and Llama-3.1-8B as interviewers and GPT-4o as the source, in three conditions: the full game, no persuasion, and no withholding (Table 2 in [1]). Five people also played the source in 72 games to check the simulation. ChatDev tests 1,200 SRDD requests against GPT-Engineer and MetaGPT, all on ChatGPT-3.5 to compare. It scores completeness, executability, consistency, and "quality", and adds GPT-4 and human preference tests.

Main findings. In both papers, giving an agent a role changes what it produces. But in NewsInterview, even the prompted interviewers rarely acknowledge answers or steer back to planned topics the way journalists do. In NewsInterview, about 0% of LLM questions are acknowledgements, against 9% for journalists, and 70% are follow-ups, against 60% (Figure 2 in [1]). The baseline LLM matches the journalist's exact question 3.9% of the time, against 8.2% for the human baseline, and matches its motivation 4.7% of the time against 35.4% (Table 1 in [1]). In the game, GPT-4o collects 50.4% of the facts, and 84.2% when sources stop withholding information (Table 2 in [1]). ChatDev scores 0.3953 on quality, against 0.1523 for MetaGPT and 0.1419 for GPT-Engineer, and 88% of its programs run, compared with 41% of MetaGPT's (Table 1 in [2]).

Purpose of the evidence. The ablations test the social part of each system. Removing ChatDev's roles drops quality from 0.3953 to 0.2212 and executability from 0.88 to 0.58 (Table 4 in [2]), so the roles, rather than the extra LLM calls, seem to drive the gains. Removing persuasion from NewsInterview[1] barely changes the scores: GPT-4o goes from 50.4% to 49.8%, and Llama-3.1-70B improves from 42.6% to 45.5% (Table 2 in [1]), which supports the claim that LLM interviewers don't persuade. Also, the human baseline sets a ceiling. If we match the exact question only 8.2% of the time, the LLM's 3.9% is far from total failure.

4. Strengths, Limitations, & Conclusions

NewsInterview's strength is natural, large, public data of 45,848 real interviews versus earlier crowdsourced or 100 to 1,000 transcript sets; that's why findings reflect professional talk. The simulated source was checked against people ($r = 0.43$), or 0.68 without adversarial personas. ChatDev's strength is component ablation: roles have the largest effect (Table 4), pointing builders toward persona design.

Limitations go past the authors' own caveats. In [1], GPT-4o judges similarity and plays every source, while GPT interviewers score highest (Table 2 in [1]); without another source model, rankings may partly reflect same-family style matching. Table 2 lacks error bars despite a stochastic game, so 1–3 point gaps are uninterpretable. Text also contradicts figures: Section 3.3 says outline prompts keep topic changes "significantly below human levels," yet Figure 2 shows 17% versus human 12%. In [2], the metrics never test whether a program does what was asked, which the authors partly admit (p. 9). Executability only means the program runs, completeness only means it has no placeholder code, and consistency barely varies (about 0.74 to 0.80), so a Gomoku game that never detects a win could still score well. GPT-4 also preferred ChatDev over MetaGPT only 57% of the time, while human judges did 88% of the time (Table 2 in [2]). Both papers rely on GPT judges with weak checks: [1]'s GPT-4o ratings were only "manually inspected," and [2]'s GPT-4 judge disagrees with humans. What the papers do show is that prompted LLM interviewers rarely acknowledge answers and extract only about half the facts, and that role prompts raise ChatDev's scores. [1]'s descriptive findings are convincing and its game results suggestive; [2]'s gains are real on its metrics, but those metrics measure if it runs, not works.

5. Follow-up Investigation

One question I would focus on: If an LLM interviewer is told to acknowledge answers and steer topics, does it get more facts, or does it only change the style? [1] found almost no acknowledgements and no gain from persuasion (Table 2 in the paper), while [2] depicted that the role prompt changes how agents will behave. So a prompt might close [1]'s grounding gap.

I am proposing to investigate the run: [1]'s public game with three interviewer prompts: (a) the original, (b) the original plus "briefly acknowledge the answer before each question", and (c) prompt (b) plus the persona tips from Table 12 in [1]. Interviews and sources stay the same, each version runs several times, and I compare the average share of facts collected, with confidence intervals. Labeling the new questions by type would confirm that acknowledgements actually increased. Because the source's scoring rules already count acknowledgement as persuasive (Appendix D.2 in [1]), I would also use a non-GPT source and repeat the human ratings. If (b) or (c) clearly beats (a), acknowledging helps get information; if not, it isn't enough on its own. The result shows whether grounding is worth building into dialogue agents, and whether my panel agents should acknowledge candidates' answers.

References

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